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Differential multimodal fusion algorithm for remote sensing object detection through multi-branch feature extraction

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ABSTRACT

Object detection through remote sensing imagery presents challenges such as a high proportion of small objects and inadequate detectability of objects in low-light environments. Despite advancements in existing methods, these challenges persist. To address them, this paper introduces MMFDet, a differential multimodal fusion algorithm for remote sensing object detection through multi-branch feature extraction. MMFDet leverages multimodal data differences and complementary features to address these issues. The proposed algorithm comprises a multi-branch feature extraction structure, a multimodal difference complement module (MDCM), and a high-level feature split hybrid module (HFSHM). First, the multi-branch feature extraction structure extracts features from different modalities, adapting to small objects through branch structure adjustments. Second, the MDCM leverages inter-modality differences to enhance the sensitivity to complementary features, addressing low-light detection challenges. Additionally, the high-level feature split hybrid module improves small object detection accuracy by splitting and hybridizing multimodal features, enabling enhanced feature integration. Experiments conducted on the VEDAI and Drone Vehicle datasets demonstrate a 14.7% and 11.1% average precision improvement over the baseline algorithm, respectively. Furthermore, compared to traditional and other multimodal remote sensing object detection algorithms, MMFDet achieves significantly superior average precision.

1. Introduction

Remote sensing image object detection represents a fundamental endeavor within the realms of remote sensing and computer vision, aimed at locating and categorizing objects of interest within remote sensing images. It plays a crucial role in applications such as environmental monitoring, disaster detection, and intelligence reconnaissance (Gao et al., 2024; Sagar et al., 2024).

In recent years, the advent of deep learning has promoted a profusion of methodologies for object detection in remote sensing images. These approaches predominantly center around single-modality image analysis, leveraging either one-stage networks typified by the YOLO series or two-stage networks exemplified by Faster R-CNN as foundational architectures to address the complexities inherent in remote sensing image object detection. Zhang et al. (2024) introduced a high-resolution feature generator tailored for detecting small objects within remote sensing images. By harnessing genuine high-resolution features to extract pertinent information from low-resolution images, this method significantly enhances recognition performance for diminutive objects. Yu and Da (2023) devised a phase-shifting encoder (PSC) to address the challenge of arbitrary object rotation angles in remote sensing images, effectively circumventing issues such as boundary discontinuity and square-like in rotated object detection. Empirical evaluations conducted on datasets such as DOTA (Xia et al., 2018) have demonstrated the superior accuracy achieved by this approach. Fu et al. (2023) advanced a pioneering anchor-free algorithm that directly predicts object positions and categories based on individual pixels within the image. This innovation facilitates the network's adeptness at accommodating objects of varying scales. Notably, this method attains commendable performance on the DIOR dataset (Li et al., 2020). Zhao et al. (2023) proposed a multi-scale attention feature fusion module predicated on the YOLOX algorithm (Ge et al., 2021). By integrating multi-scale contextual and channel information, rather than applying simplistic summation and concatenation operations in feature fusion, this methodology mitigates the encumbrance posed by intricate background information. Such an

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approach effectively addresses challenges arising from significant scale discrepancies, intricate background contexts, and densely distributed objects in remote sensing images. Despite the considerable strides made in enhancing the accuracy of object detection in remote sensing images, identifying objects in low-light conditions is highly challenging. Single-modality information alone is insufficient in surmounting this obstacle and fails to propel object detection in remote sensing images beyond its current threshold.

With the progression of remote sensing imaging technology, the volume of multimodal remote sensing imagery data continues to escalate, affording researchers ready access to multimodal remote sensing data replete with complementary information (Li, Hong et al., 2022). A multitude of scholars are dedicated to exploring the realm of multimodal information fusion. Tang et al. (2022) introduced a light perception network to steer the training of the fusion network, thereby seamlessly integrating the complementary information emanating from visible and infrared modalities contingent upon lighting conditions, thus engendering higher quality images. Liu et al. (2022) employed a target-aware dual adversarial learning network for multimodal image fusion, adept at discerning commonalities amidst disparities and synthesizing structural information and textural details from two modalities. This model adeptly adapts to downstream tasks. Ma et al. (2022) introduced the Swin Transformer (Liu et al., 2021) to construct intra-domain and cross-domain fusion units, thereby integrating intra-domain dependencies and cross-domain long-distance dependencies. By thoroughly extracting differential and complementary information from each modality, the network effectively fuses multimodal information within the extracted feature maps.

In existing studies, objects with a resolution of less than 32 × 32 pixels, or those with a median ratio of the area of the bounding box to the image area of all objects in the same category between 0.08% and 0.58%, are usually defined as small objects. Due to the low resolution and inconspicuous features of small objects, they are easily interfered by background noise in the image, resulting in difficulties in localizing and identifying small objects. Although many algorithms have been proposed for small object detection, these algorithms often use complex structural designs to improve the accuracy of small object detection, resulting in high model complexity. Under low-light conditions, existing methods often use Generative Adversarial Network (GAN)-based methods to enhance the image, making the object more prominent. However, an image enhancement network needs to be added before the object detection network, which undoubtedly adds the complexity of the network. Mild low-light in remote sensing image object detection results in reduced image details. Moderate low-light and heavy low-light have increased image noise and loss of detail information severely, making it almost impossible to detect objects. To the best of our knowledge, few studies have addressed the problem of difficult object detection in low-light remote sensing images through a multimodality approach.

The low accuracy of small object detection and the difficulty of detecting objects in low-light conditions have become important issues that restrict the development of remote sensing image object detection. Therefore, in this paper, we hope to solve the problem of low accuracy of small object detection in remote sensing image object detection by designing the network structure as simple as possible. And the multimodality image is used as input to overcome the problem of difficult detection of objects in low-light images by utilizing the difference information between different modalities. First, this study established a multi-branch feature extraction structure, composed of primary branches and an auxiliary branch, designed for extracting features from RGB images, IR images, and the pixel-level fusion of the two image modalities, respectively. Subsequently, within the context of feature extraction utilizing the multi-branch framework, a multimodal difference complement module was employed to address the challenge of detecting objects in low-light environments within RGB images by leveraging the disparities between RGB and IR images. Specifically, the IR image features complemented those of the RGB images. Finally, this paper proposes the incorporation of a high-level feature split hybrid module to handle the input of the feature fusion layer. This module encompasses a max pooling channel, an attention channel, and a convolutional channel, which extracts local detail information, attention feature information, and global feature information, respectively. This facilitates the amalgamation of a broader array of information conducive to small object detection within the feature fusion layer. The main contributions of this study are as follows:

(1) A multi-branch feature extraction structure was proposed. In the single-modality feature extraction branches, unlike traditional methods that use larger stride, the loss of detail information was avoided by designing appropriate stride parameters, which makes it suitable for small object detection utilizing such a simple design. Moreover, additional information complementation was provided by introducing auxiliary branching to retain the model's ability to detect objects of other sizes.

(2) A multimodal difference complement module was constructed to overcome the problem of difficult object detection in low-light images by fusing features between multimodal images. This module was applied to each stage of feature extraction. Different from the simple concatenation or summation of features by existing methods, this module subtly utilized the difference information between modalities, amplifies that difference information and performs feature fusion. The features between different modalities were made fully complementary, which effectively improves the detection accuracy in low-light images.

(3) A high-level feature split hybrid module was developed to operate on the input features of the feature fusion layer. Different from the traditional methods that do nothing to the input features of the feature fusion layer. The high-level feature split hybrid module proposed in this paper performs operations such as splitting, multi-channel extraction, and hybridizing on the input features to highlight the key features of small objects and reduce the interference of background noise on small object detection.

2. Related work

2.1. Multimodal remote sensing tasks

With the advancement of deep learning, single-modality data has emerged as a constraining factor in further enhancing tasks within the remote sensing domain. Many challenges in these tasks cannot be adequately addressed solely through single-modality data. Therefore, utilizing multimodal data to address problems pertaining to remote sensing tasks has garnered significant research attention in recent years.

Within the realm of remote sensing, diverse modalities of data exist, including Red–Green–Blue (RGB), infrared (IR), hyperspectral, multispectral, and Synthetic Aperture Radar (SAR). These multimodal datasets from varied sensors can mutually complement one another, thereby enhancing the performance of diverse tasks (Barbato et al., 2024; Sun et al., 2023). For example, RGB images often exhibit diminished visibility in low-light conditions, leading to diminished accuracy in object detection. However, infrared images can effectively compensate for this limitation of RGB images.

Wang et al. (2023) proposed a multi-layer cross-Transformer to facilitate feature interaction between hyperspectral and multispectral images at equivalent scales, enabling seamless integration of spectral and spatial information to augment the accuracy of subsequent recognition and classification tasks. Kieu et al. (2023) introduced a spatial and volumetric learning component based on the PerceiverIO architecture, aiming to achieve more accurate semantic segmentation of remote sensing images by harnessing the complementary information from hyperspectral images and LiDAR DSM. Xu et al. (2024) delineated a causal feature extraction framework by incorporating causal inference, thereby effectively enhancing the accuracy of multimodal pixel-level classification. The efficacy of this approach was comprehensively validated across hyperspectral-SAR and hyperspectral-LiDAR

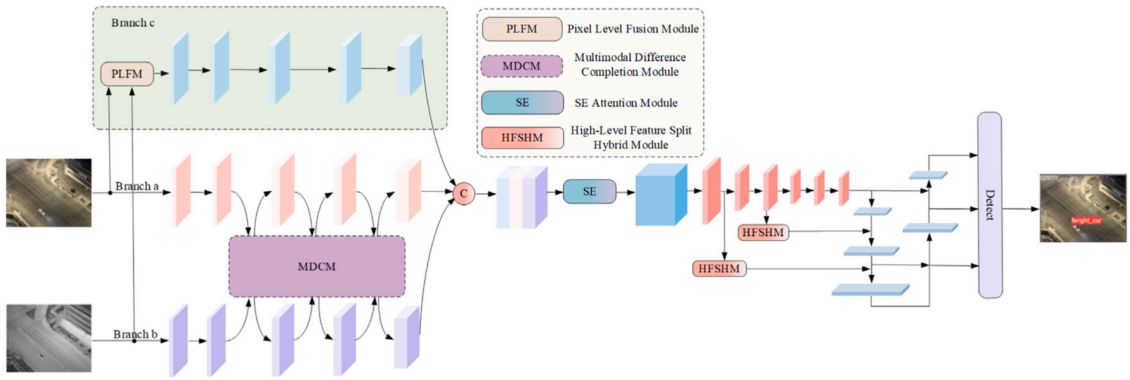

Fig. 1. Overall structure of the MMFDet algorithm.

datasets. Ye et al. (2024) proposed a saliency feature fusion framework to reconstruct the visual saliency feature map through fusion of the complementary features derived from optical and SAR images. This framework adeptly mitigates the adverse effects stemming from modality discrepancies and scattering noise, thereby yielding superior outcomes. Li, Li et al. (2022) devised a spatio-temporal fusion method to mitigate the disparities arising from distinct imaging mechanisms between SAR and optical sensors. This method adeptly amalgamates SAR images with optical images to facilitate high-precision reconstruction of the normalized difference vegetation index (NDVI).

2.2. Multimodal remote sensing image object detection

Leveraging complementary information from diverse modalities is crucial for enhancing the accuracy of object detection in remote sensing images. Existing multimodal fusion techniques are primarily categorized into pixel-level fusion, feature-level fusion, and decision-level fusion (Gómez-Chova et al., 2015). Pixel-level fusion entails image fusion at the lowest level. While this method necessitates less computation, it often yields suboptimal fusion outcomes and proves challenging to implement with unaligned multimodal images. Decision-level fusion integrates the decision outcomes of distinct modalities at the final stage. However, such approaches typically entail substantial redundant computation across modal branches, leading to heightened computational complexity and slower model speed. Therefore, feature-level fusion methods are commonly employed for object detection in multimodal remote sensing images.

Zhang et al. (2023) employed a pixel-level fusion approach to amalgamate information from RGB and IR modalities, introducing a super-resolution branch to generate high-resolution features during the training phase, which was subsequently discarded during inference. Experiments conducted on the VEDAI dataset (Razakarivony & Jurie, 2016) evinced the superior performance of this method compared to object detection tasks performed on single-modality images. Gu and Hu (2022) devised a balanced multimodal feature fusion technique grounded in the YOLOv4 algorithm (Bochkovskiy et al., 2020). By executing feature-level fusion, this method adeptly integrates information from RGB and IR images, thereby markedly enhancing the detection accuracy of small objects in multimodal scenarios. Sharma et al. (2020) proposed YOLOrs, predicated on the ResNet (He et al., 2016) and YOLOv3 (Redmon & Farhadi, 2018) architectures. This approach introduces a novel mid-level fusion architecture, effecting weighted combinations between the outputs of two distinct encoders catering to different modalities of remote sensing images. Tian et al. (2023) harnessed both convolutional neural networks and Transformers to extract features from diverse modalities, merging the local features from RGB images with the global features from IR images. Additionally, they proposed an adaptive multimodal feature fusion module to seamlessly integrate features from the two modalities, thereby enhancing the accuracy of object detection in remote sensing images.

2.3. Small object detection

Although object detection algorithms are continuously improved and optimized, these conventional algorithms still show a large number of missed detections when detecting small objects. In order to solve the problems of small object detection, such as susceptibility to background noise interference and dense arrangement, the research for small object detection has gradually become the focus of the object detection field.

Qi et al. (2022) proposed a single-stage small object detection algorithm for the problems of insufficient spatial information of the target object and poor fusion of semantic and spatial information. The accuracy of small object detection is effectively improved by adaptive spatial parallel convolution as well as multi-scale fusion module. Zhu et al. (2024) proposed a small object detection method based on global multilayer sensing and dynamic region aggregation, which better solved the problems of dense distribution of small objects and noise interference. Li et al. (2024) proposed a global and local attention mechanism for the small and densely distributed objects in remote sensing images. Combining local features with global feature processing improves the performance of small object detection in remote sensing image. Zhu et al. (2023) proposed a small object detection algorithm for driver distraction detection. The algorithm improves the multi-dimensional feature extraction capability by fusing multi-dimensional adaptive attention and convolutional neural network, which effectively improves the detection performance of tiny objects.

The above methods can better solve the problems of small object detection, such as the dense arrangement and the susceptibility of small objects to background noise interference. However, the existing studies do not consider improving small object detection accuracy by designing the network structure as simple as possible, and there is a lack of research on small object detection in low-light images.

3. Method

The proposed MMFDet network structure is depicted in Fig. 1. The MMFDet algorithm primarily comprises three core modules: a multi-branch feature extraction structure, a multimodal difference complement module, and a high-level feature split hybrid module. This section discusses network architectures and module specifications of MMFDet.

3.1. Multi-branch feature extraction structure

Given the substantial complementarity between RGB and IR images, the challenges in multimodal remote sensing image object detection revolve around fully integrating information from different modalities, leveraging this complementary information to detect objects in low-light environments, and enhancing the detection accuracy of small objects through judicious network design. To address this challenge, we

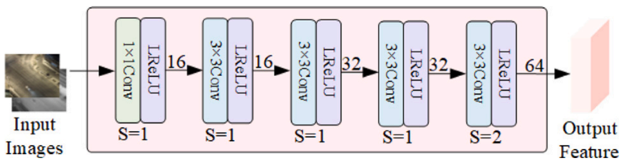

Fig. 2. Single modality feature extraction branch structure.

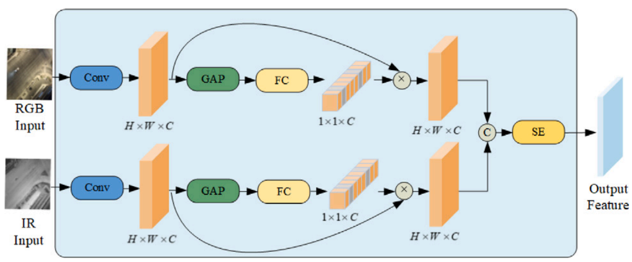

Fig. 3. Pixel-level fusion structure.

have devised a multi-branch feature extraction structure, illustrated in Fig. 1. It comprises three branches in total: branches a and b serve as the main branches, responsible for extracting feature information from the RGB and IR modalities, respectively, while branch c functions as an auxiliary branch, managing the feature information derived from the pixel-level fusion of RGB and IR images.

Branch a receives the RGB image as input, while branch b receives the IR image. The fusion of multimodal information between branches a and b employs a feature-level fusion approach, elucidated in detail in Section 3.2. The network architectures of branches a and b are identical, as depicted in Fig. 2. Each branch comprises five convolutional layers. Initially, a 1 × 1 convolution is employed to mitigate modality differences between the RGB and IR images, ensuring alignment of their channel counts. Subsequently, three 3 × 3 convolutions are utilized to extract features from both the RGB and IR images. Notably, the stride of these three convolutional layers is set to 1, facilitating extraction of more contextual information while minimizing loss of detailed information. This design is particularly suitable for remote sensing images containing numerous small objects. The final convolutional layer employs a 3 × 3 convolution with a stride of 2 to conduct deep feature extraction, reducing the size of the feature map and thus alleviating computational burden to prevent excessive inference time. All layers in branches a and b adopt Leaky ReLU as the activation function. The size of the convolution kernel used in branches a and b is the same as the conventional method, with a size of 3 × 3. The stride of the convolution kernel is adopted with a smaller value, which is designed to be more suitable for small object detection. This is because a smaller stride can try to ensure the high resolution of the feature map as much as possible, which helps to capture the detail information. At the same time, a smaller stride can avoid missing some pixels when the convolution kernel is sliding, thus reducing the loss of information related to small objects.

To provide additional information complement to the single-modality branches and preserve the model's ability to detect objects of other sizes, we introduce an auxiliary branch, denoted as branch c. Branch c takes the pixel-level fusion of the RGB image and the IR image as its input. The pixel-level fusion belongs to the low-level fusion and introduces less computation compared to other feature-fusion methods, so we choose pixel-level fusion as an auxiliary branch. The structure of the pixel-level fusion module illustrated in Fig. 3. Initially, a convolution operation is applied to increase the channel dimensions of both the RGB image and the IR image. Subsequently, an adaptive global average pooling operation and a fully connected layer compress the spatial dimensions, generating channel weights.

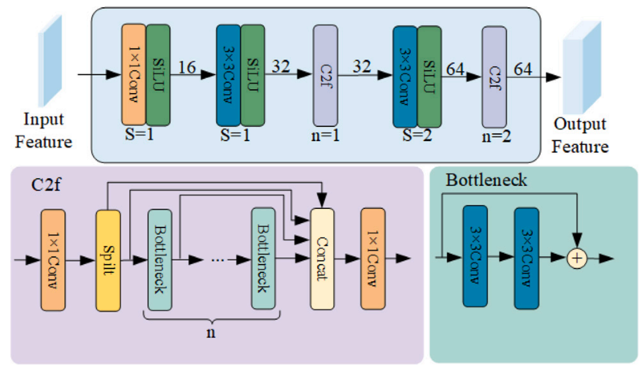

Fig. 4. Auxiliary branch feature extraction structure.

This adaptive mechanism enables the network to learn the significance of information from each modality dynamically. The channel weights of the two branches are then multiplied with their respective features and concatenated. Finally, the SE attention module (Hu et al., 2018) is employed to produce the fused features. In the subsequent feature extraction process, the fused image features undergo convolutional operations and cross-stage partial bottleneck fusion modules. The structure of this feature extraction process is delineated in Fig. 4.

The feature map output after pixel-level fusion is first passed through a 1 × 1 convolution when input into the feature extraction structure. This convolution enables pixel-level cross-channel information interaction at the lower layers and maps the information into a high-dimensional space. Subsequently, a 3 × 3 convolution is employed for deep feature extraction. Similarly, the stride of this convolutional operation is set to 1 to extract more detailed information that is beneficial for small object detection. In the third layer, the feature map passes through a C2f module, which does not alter the channel size but is solely intended to enhance gradient echo. Subsequently, the feature map undergoes a 3 × 3 convolution operation with a stride of 2, reducing the size of the feature map to avoid excessive computational burden in the auxiliary branch. Finally, the feature map is output after passing through a C2f module.

The details of all the branches in the multi-branch feature extraction structure such as the name of the module, the size of the convolutional kernel, the configuration parameters of the module and the number of output channels are shown in Table 1.

The role of branch a and branch b is to extract the features of the single-modality image, focusing on the extraction of detailed features in the process of feature extraction to reduce the loss of information related to small objects. Because branch c does not design many convolutional modules with stride 1 as in branches a and b. Instead, it utilizes the C2f module for feature extraction, which aggregates features of different scales. Therefore the role of branch c is to extract the features of other sized objects to provide additional information complement for the model, preserving the model's ability to detect other sized objects.

As depicted in Fig. 1, the output feature maps from the three branches are concatenated along the channel dimension. Subsequently, a SE channel attention module is applied to assign varying weights to the feature maps extracted from the main and auxiliary branches. This yields the final output feature map for subsequent feature extraction.

3.2. Multimodal difference complement module

The RGB image encapsulates rich feature information such as texture and contour. However, recognizing object features in low-light environments proves challenging. Conversely, the IR image, derived from measuring the heat radiation of objects, remains unaffected by light conditions and effectively compensates for the deficiencies of the RGB image. Consequently, in this study, IR image features are

Table 1
Details of all the branches in the multi-branch feature extraction structure.

	Branches a and b				Branch c			
	Module name	Kernel size	Configuration	Output channels	Module name	Kernel size	Configuration	Output channels
Layer1	Conv	1 × 1	Stride = 1	16	Conv	1 × 1	Stride = 1	16
Layer2	Conv	3 × 3	Stride = 1	16	Conv	3 × 3	Stride = 1	32
Layer3	Conv	3 × 3	Stride = 1	32	C2f		n = 1	32
Layer4	Conv	3 × 3	Stride = 1	32	Conv	3 × 3	Stride = 2	64
Layer5	Conv	3 × 3	Stride = 2	64	C2f		n = 2	64

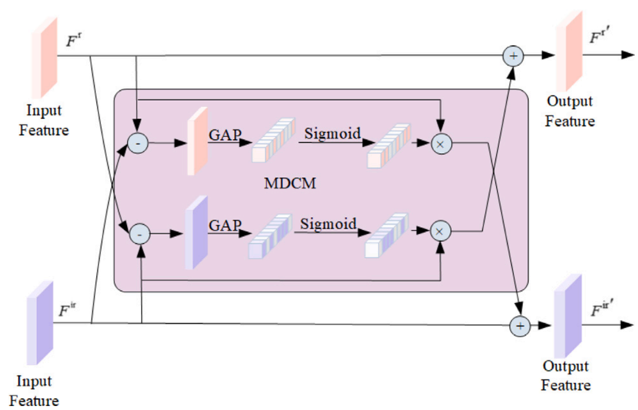

Fig. 5. Multimodal difference complement module structure.

employed as a supplement to RGB images to address the difficulty of detecting objects in low-light environments.

To efficiently leverage the differential information across different modalities, we draw from Zhou et al. (2020) to propose a multimodal difference complement module aimed at enhancing the absorption of information from each modality to the other. Traditional methods involving direct concatenation or addition of feature maps often fail to effectively emphasize the differential information between different modalities, resulting in inadequate utilization of multimodal complementary information and limited enhancement in detection accuracy. Therefore, this paper presents an effective and straightforward approach to multimodal fusion. The multimodal difference complement module utilizes the differential information between the two modalities to bolster detection accuracy through cross-modality complementarity.

First, we denote the IR feature map as F^{ir} and the RGB feature map as F^r. Both F^{ir} and F^r can be represented by their respective modality similarities and complementarities as follows:

$$F^{ir} = \frac{F^{ir}+F^{ir}}{2} + \frac{F^r - F^r}{2} = \frac{F^r+F^{ir}}{2} + \frac{F^{ir}-F^r}{2} \tag{1}$$

$$F^{r} = \frac{F^{r}+F^{r}}{2} + \frac{F^{ir} - F^{ir}}{2} = \frac{F^r+F^{ir}}{2} + \frac{F^{r}-F^{ir}}{2} \tag{2}$$

From Eqs. (1) and (2), it can be seen that the features of two different modalities are both composed of two parts. One part is the same, which we call the similar part, i.e. $\frac{F^r+F^{ir}}{2}$. The other part is different, that is, each modality possesses different characteristics, which we call the complementary part. We borrow the idea of suppressing the common mode signal and amplifying the differential mode signal as in the case of differential amplifier circuits. We hope that each modality can retain the characteristics of the similar part, and based on this, amplify the complementary characteristics with differences. So that one modality learns the differential information from the complementary features of the other modality and improves the sensitivity to the other modality. The structure of the MDCM is shown in Fig. 5, and the main idea is to obtain complementary features from another modality through differential weighting of channel dimensions.

The input to the MDCM is the feature map obtained during the feature extraction process of the single-modality branch. The computational process of this module is elucidated in Eqs. (3) and (4):

$$F^{ir'} = F^{ir} + \delta(GAP(F^r - F^{ir})) \odot F^r \tag{3}$$

$$F^{r'} = F^{r} + \delta(GAP(F^{ir} - F^{r})) \odot F^{ir} \tag{4}$$

Here, $\odot$ represents element-wise multiplication, $\delta(\cdot)$ denotes the sigmoid activation function, and $GAP(\cdot)$ represents adaptive global average pooling. As illustrated in Eqs. (3) and (4), the MDCM first performs a subtraction operation on the features of the two modalities to obtain the difference features. Next, the difference features are compressed into a global difference vector containing channel-wise difference statistics using adaptive global average pooling. The global difference vector is able to represent the difference between RGB and IR modalities. The fusion weight vector is then obtained by using a Sigmoid activation function on the global difference vector. Next F^r and F^{ir} are multiplied with their respective obtained fusion weight vectors and added with the features of the original modality as complementary feature information to obtain the output features. The output feature maps of the multimodality difference complement module are $F^{r'}$ and $F^{ir'}$, respectively. As shown in Fig. 1, the outputs of this module are connected to the respective single-modality feature extraction branches for subsequent feature extraction.

The feature map processed by the MDCM module fully combines the feature information of texture and contour contained in the RGB modality and the thermal radiation information of the object contained in the IR modality. For RGB images with low lighting conditions, the thermal radiation information of the object in IR modality can be used as a supplement to effectively improve the detection in low-light environments.

3.3. High-level feature split hybrid module

To enhance the detection accuracy of small objects in remote sensing images, we introduce a high-level feature split hybrid module, as depicted in Fig. 6.

The high-level feature split hybrid module accepts the mixed features extracted from the feature extraction network as input, and the output features processed by this module are utilized in the feature fusion layer. This module encompasses multiple branches. Upon entering the high-level feature split hybrid module, the input feature map undergoes division into three segments, which are subsequently fed into the max pooling channel, attention channel, and convolution channel, respectively. Assuming an input feature map $X \in \mathbb{R}^{H\times W\times C}$, it is split along the channel dimension into $X_p \in \mathbb{R}^{H\times W\times C_p}$, $X_a \in \mathbb{R}^{H\times W\times C_a}$, and $X_d \in \mathbb{R}^{H\times W\times C_d}$, where $C = C_p + C_a + C_d$. Subsequently, X_p, X_a, and X_d represent inputs to the max pooling channel, attention channel, and convolution channel, respectively, for the extraction of local detail information, attention feature information, and global feature information.

As depicted in Fig. 6, the max pooling channel comprises a cascaded max pooling layer and a linear layer. The attention channel integrates a hybrid attention mechanism, inspired by the attention mechanism proposed in Woo et al. (2018). The convolution channel includes a

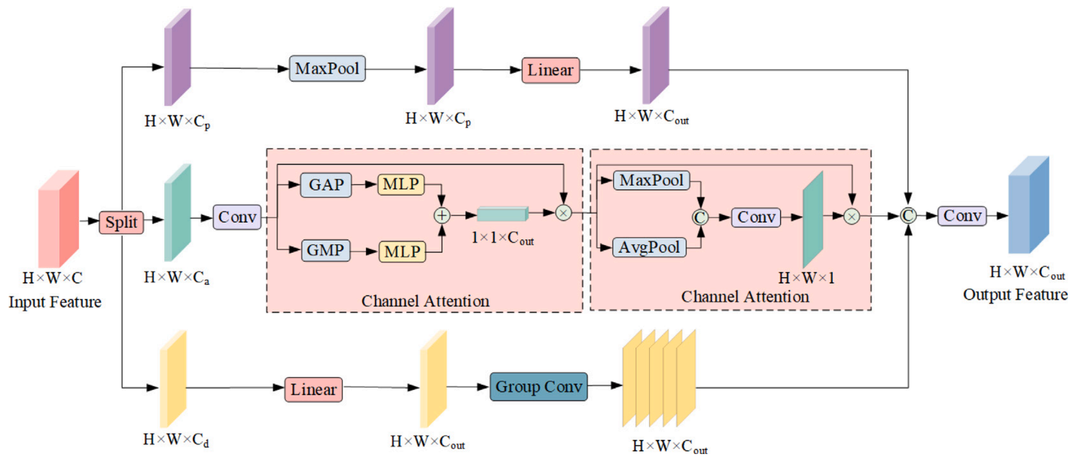

Fig. 6. High-level feature split hybrid module structure.

linear layer and a group convolution. The computational processes of these three channels are represented by Eqs. (5), (6), and (7), respectively.

$$O_p = FC(MaxPool(X_p)) \tag{5}$$

$$O_a = Attention(X_a) \tag{6}$$

$$O_d = GroupConv(FC(X_d)) \tag{7}$$

Here, O_p, O_a, and O_d denote the outputs of the max pooling channel, attention channel, and convolution channel, respectively. $Attention(\cdot)$ represents the processing of the hybrid attention mechanism. The hybrid attention mechanism comprises channel attention and spatial attention. Upon entering the attention channel, the input feature map undergoes channel attention, which involves adaptive global average pooling and adaptive global max pooling to compress the width and height of the feature map. Subsequently, the compressed features are passed through the multilayer perceptron and added, followed by their normalization using the sigmoid activation function to obtain channel attention weights. These weights are then multiplied with the input to produce the channel attention feature map, which serves as the input for the spatial attention layer. The spatial attention layer performs max pooling and average pooling operations on the input along the channel dimension, followed by concatenation and a convolution operation to reduce the number of channels to 1. Spatial attention weights are generated using the sigmoid activation function and multiplied with the input to produce the hybrid attention feature map.

The output of the high-level feature split hybrid module is obtained by concatenating the outputs of the three branches along the channel dimension and adjusting the number of channels through a convolution operation, as shown in Eq. (8):

$$O = Conv(Concat(O_p, O_a, O_d)) \tag{8}$$

As depicted in Fig. 1, the high-level feature split hybrid module is employed twice before the feature fusion layer to process the input. Existing methods usually do not process the inputs of the feature fusion layer and only perform simple concatenation. Instead, the high-level feature split hybrid module splits the high-level features extracted by the feature extraction network in three equal parts along the channel dimensions, and inputs them into different channels for processing. A portion of these features go through the convolutional channel. This channel utilizes linear layers with group convolution to further extract features. The output features are made to contain larger scale features to avoid the degradation of the network's ability to detect objects of other scales. The second part of the features goes through the max pooling channel. This channel retains the salient features of small objects in the feature map through the max pooling operation. And suppresses the background noise to some extent, since the noise is usually not the maximum value in the local region. Thus generating features that are more favorable for small object detection. The third part of the features goes through the attention channel. This channel utilizes the hybrid attention mechanism to adaptively adjust the model's attention distribution. This channel can highlight the key features and suppress the interference of background information, thus enhancing the features of small objects and reducing the interference of background noise on small object detection. Finally, the output features of the three channels are hybridized by concatenation and convolution operations and fed into the feature fusion layer. Thus, the feature fusion layer fuses more key features that are helpful for small object detection and improves the accuracy of MMFDet for small object detection.

4. Experiments and results

4.1. Dataset and evaluation metrics

4.1.1. Dataset

This experiment utilizes the VEDAI dataset and the Drone Vehicle dataset (Sun et al., 2022) as the experimental datasets.

The VEDAI dataset comprises 1246 images, each containing two modalities: RGB and IR. Typically, instances in this dataset represent small objects with limited pixel points. After filtering, we obtained a total of 1210 images, of which 1089 images were randomly selected for the training set, and the remaining 121 images constituted the test set.

To further demonstrate the ability of the proposed algorithm to detect objects in low-light environments, we selected the Drone Vehicle dataset as the second dataset. This dataset comprises 28,439 images, all containing both RGB and IR modalities, and includes a substantial number of images captured in low-light conditions. It encompasses five categories: car, truck, bus, van, and freight car. The training set comprises 17,990 images, the validation set contains 1469 images, and the test set includes 8980 images. In the Drone Vehicle dataset, instances are annotated using rotated bounding boxes represented by the coordinates of four points in the form of $(x_1, y_1; x_2, y_2; x_3, y_3; x_4, y_4)$. To standardize the dataset representation, we convert these rotated boxes into horizontal bounding boxes in the form of (x_c, y_c, w, h), where x_c, y_c, w, h represent the horizontal and vertical coordinates of the center point and the width and height of horizontal bounding boxes, respectively.

$$x_c = \frac{max(x_1, x_2, x_3, x_4) + min(x_1, x_2, x_3, x_4)}{2 \times w_i} \tag{9}$$

Table 2
Ablation studies on the VEDAI dataset.

Method	mAP/%
Baseline	63.2
Baseline+MDCM	74.0
Baseline+HFSHM	72.1
Baseline+MDCM+HFSHM	76.7
Baseline+MDCM+MBE	74.6
Baseline+MDCM+MBE+HFSHM	**77.9**

$$y_c = \frac{max(y_1, y_2, y_3, y_4) + min(y_1, y_2, y_3, y_4)}{2 \times h_i} \tag{10}$$

$$w = \frac{max(x_1, x_2, x_3, x_4) - min(x_1, x_2, x_3, x_4)}{w_i} \tag{11}$$

$$h = \frac{max(y_1, y_2, y_3, y_4) - min(y_1, y_2, y_3, y_4)}{h_i} \tag{12}$$

where w_i and h_i represent the width and height of the image, respectively.

4.1.2. Evaluation metrics

In this study, the following metrics are employed to evaluate performance: Average Precision (AP), mean Average Precision (mAP), and False Negative Rate (FNR) for accuracy evaluation, and Parameters and GFLOPs for model size evaluation.

4.2. Experimental environment and parameter settings

This experiment was conducted on the Windows 11 operating system using an NVIDIA GeForce RTX 4070 GPU. The deep learning framework utilized was PyTorch 2.0.1, with CUDA version 11.8. Stochastic Gradient Descent (SGD) was employed for optimization, with a total of 300 epochs of training. The initial learning rate for this experiment was set to 0.01, with a minimum learning rate of 0.001 and a batch size of 2. Warm-up training of 3 epochs was conducted at the beginning of training. During the warm-up training phase, the momentum parameter was set to 0.8 and adjusted to 0.937 after completing the warm-up training.

4.3. Experimental results and analysis

4.3.1. VEDAI

In this study, six ablation experiments were conducted on the VEDAI dataset to illustrate the effectiveness of the proposed modules. The experimental results on the VEDAI dataset are summarized in Table 2.

Because the VEDAI dataset primarily consists of small objects and does not include images captured in low-light environments, except for the MMFDet algorithm proposed in this paper, other algorithms adopt pixel-level fusion methods, as compared in subsequent sections. As observed from Table 2, adding the MDCM module improves the mAP value by 10.8% compared to the baseline algorithm. Further, incorporating the HFSHM module enhances the mAP value by 8.9%. Integrating the MDCM module and HFSHM module leads to a 13.5% improvement in the mAP value compared to the baseline algorithm. Additionally, integrating the MDCM module with the MBE structure boosts the mAP value by 11.4% over the baseline algorithm. Finally, employing all the improved methods together enhances the mAP by 14.7%.

To validate the effectiveness of the proposed MDCM module for multimodal information fusion, we conducted comparative experiments using different input modes and fusion methods, all based on the YOLOv8 algorithm. The results are presented in Table 3.

As indicated by Table 3 and Fig. 7, pixel-level fusion involves directly concatenating images of different modalities, improving the mAP value by 3.3% and 12% over single-modality RGB and IR images, respectively. However, feature-level fusion, which concatenates features of two modalities at different stages of feature extraction, yields mAP values higher than those from single-modality IR images. Notably, the mAP values decrease by 6.3%, 3.3%, 3.4%, and 2.1%, respectively, compared to single-modality RGB images, indicating that simple concatenation of features is not an optimal multimodal fusion method. Conversely, adopting the MDCM fusion method proposed in this paper results in mAP values increasing by 14.1% and 22.8% over single-modality RGB and IR images, respectively, surpassing all other multimodal fusion methods. This underscores the effectiveness of the MDCM fusion method in fusing complementary information between different modalities and enhancing object detection accuracy in remote sensing images.

In order to evaluate the effect of different fusion methods as auxiliary branches on the model complexity, we designed five sets of experiments with different fusion methods, and the experimental results are shown in Table 4. The pixel-level fusion method has the smallest GFLOPs and number of parameters. In addition, as shown in Table 3, the mAP values of pixel-level fusion methods are also much higher than those of feature-level fusion. Therefore, we choose pixel-level fusion as an auxiliary branch, which can improve the detection accuracy while reducing the model complexity.

In this paper, the convolutional stride parameters of branches a and b in the multi-branch feature extraction structure are adjusted. The relevant parameters of the branch c are consistent with the conventional method, so as to ensure that the output features are aligned with the output features of a and b branches. In order to obtain the most suitable stride parameters for small object detection, we set up four sets of experiments. In order to reduce the influence of auxiliary branch on the experimental results, we remove the auxiliary branch in this experiment, and the feature extraction structure only consists of single-modality branch a and branch b. The feature fusion method uses MDCM. The results are shown in Table 5.

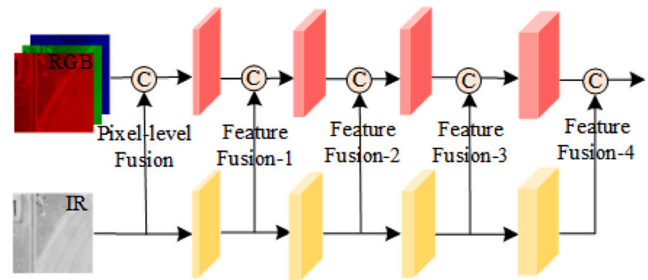

Fig. 7. Multimodal fusion methods.

Table 3
Comparison of detection results of different fusion methods on VEDAI dataset.

Fusion method	mAP/%
Single RGB	59.9
Single IR	51.2
Pixel-level Fusion	63.2
Feature-level Fusion-1	53.6
Feature-level Fusion-2	56.6
Feature-level Fusion-3	56.5
Feature-level Fusion-4	57.8
MDCM	**74.0**

Table 4
Performance comparison of different fusion methods.

Method		GFLOPs	Parameters
Feature-level Fusion	Fusion-1	9.6	3.0170M
	Fusion-2	9.9	3.0134M
	Fusion-3	12.0	3.3084M
	Fusion-4	12.3	3.0165M
Pixel-level fusion		**8.2**	**3.0125M**

Table 5
Comparison of different stride settings.

Method	mAP/%	GFLOPs
A	62.4	**14.4**
B	68.8	46.2
C	74.0	150.6
D	**79.7**	522.9

Table 6
Comparison of detection results of different algorithms on VEDAI dataset.

Method	mAP/%	FNR/%	Params.
YOLOv3	61.3	37.2	61.5M
YOLOv4	62.5	43.3	52.5M
YOLOv5s	56.8	48.9	7.1M
YOLOv8n	63.2	37.7	**3.0M**
YOLOv8s	59.7	39.7	11.1M
YOLOv8m	62.2	35.9	25.8M
YOLOv8l	63.1	33.7	43.6M
YOLOv8x	67.3	37.3	68.1M
YOLOrs	59.2	37.7	20.2M
SuperYOLO	67.7	31.0	4.8M
MMFDet	**77.9**	**29.8**	3.5M

In Table 5, methods A, B, C, and D represent the first two convolutional kernels with stride 1, the first three convolutional kernels with stride 1, the first four convolutional kernels with stride 1, and all the five convolutional kernels with stride 1. As can be seen from Table 5, although the mAP value is optimal when the stride of all five convolutional kernels is set to 1, its GFLOPs is as high as 522.9, which makes the model computationally expensive. Therefore, we choose to set the step size of the first four convolutional kernels to 1, trying to improve the accuracy while making the model complexity small.

The effectiveness of the proposed algorithm was demonstrated by comparing it with other algorithms on the VEDAI dataset. The results are presented in Table 6.

As illustrated in Table 6, our algorithms exhibit notable improvements in mAP values compared to the YOLO series algorithms YOLOv3, YOLOv4, YOLOv5s, YOLOv8n, YOLOv8s, YOLOv8 m, YOLOv8l, and YOLOv8x. Specifically, our algorithms show enhancements of 16.6%, 15.4%, 21.1%, 14.7%, 18.2%, 15.7%, 14.8%, and 10.6%, respectively, while reducing the FNR by 7.4%, 13.5%, 19.1%, 7.9%, 9.9%, 6.1%, 3.9%, and 7.5%, respectively. Despite having slightly more parameters than YOLOv8n by 0.5M, our proposed algorithm outperforms the other seven algorithms with reductions of 58M, 49M, 3.6M, 7.6M, 22.3M, 40.1M, and 64.6M parameters, respectively. When compared to other multimodal remote sensing image object detection algorithms such as YOLOrs and SuperYOLO, our proposed algorithm achieves an improvement in mAP by 18.7% and 10.2%, respectively, along with reductions of 7.9% and 1.2% in FNR, and reductions of 16.7M and 1.3M in the number of parameters, respectively. These findings highlight significant advantages in both accuracy and model size for our algorithm.

By examining the detection accuracy of each object category in the VEDAI dataset between our proposed algorithm and other object detection algorithms, significant improvements in detection accuracy for all object categories are observed. The detailed results are presented in Table 7. The first row of Table 7 indicates the various categories. While our algorithm does not exhibit the highest detection accuracy for the Truck category, it maintains an overall superior average accuracy compared to other algorithms.

4.3.2. Drone vehicle

To assess MMFDet's efficacy in low-light conditions, we conducted six sets of ablation experiments on the Drone Vehicle dataset, with results outlined in Table 8.

Because the Drone Vehicle dataset encompasses numerous low-light images, except for multimodal remote sensing image object detection algorithms, all others were assessed solely with RGB images. Table 8 indicates that employing the MDCM fusion method yields a 10.4% improvement in mAP compared to the baseline, showcasing its ability to leverage complementary information between RGB and IR images, thereby enhancing detection in low-light settings. Integrating the HFSHM module enhances mAP by 0.4% when added to the baseline. Combining MDCM and HFSHM boosts mAP by 10.7% over the baseline. Further, when MDCM and MBE are applied together, the mAP improves by 11.0% over the baseline and 0.6% over MDCM alone. Implementing all proposed enhancements collectively enhances mAP by 11.1% compared to the baseline.

We compared the performance of different algorithms on the Drone Vehicle dataset, as detailed in Table 9.

As depicted in Table 9, compared to traditional object detection algorithms such as YOLOv3, YOLOv4, YOLOv5s, YOLOv8n, YOLOv8s, YOLOv8 m, YOLOv8l, and YOLOv8x, our algorithms demonstrate improvements in mAP values by 11.3%, 10.0%, 11.9%, 11.1%, 8.3%, 6.6%, 6.1%, and 6.3%, respectively. Moreover, the FNR is reduced by 8.9%, 8.4%, 11.5%, 10.8%, 8.4%, 7.3%, 5.4%, and 7.1%, respectively. Despite our algorithm not being optimal in terms of parameter count, both mAP value and FNR exhibit significant enhancements compared to other algorithms. In comparison with other multimodal remote sensing image object detection algorithms, YOLOrs and SuperYOLO, MMFDet exhibits a 7.1% and 4.6% improvement in mAP value, and a 0.9% and 1.1% reduction in FNR, respectively, representing a substantial advantage. Furthermore, when evaluating the MMFDet algorithm against various object detection algorithms on the Drone Vehicle dataset, it maintains the highest detection accuracy across all categories, with notably superior average accuracy. Table 10 provides a detailed comparison of detection results among different algorithms in the Drone Vehicle dataset.

As illustrated in Table 10, the MMFDet algorithm consistently achieves the highest detection accuracy for all five categories in the Drone Vehicle dataset, significantly outperforming other algorithms. Despite the slight increase in model size compared to the YOLOv8n algorithm, MMFDet maintains a significant advantage in model size compared to other algorithms.

In order to further validate the detection ability of the MMFDet algorithm under low-light conditions, we conducted experimental comparisons under normal light conditions and low-light conditions. We divide the test set of the Drone Vehicle dataset into a normal light conditions test set and a low-light conditions test set according to the light conditions, and test the model on these two test sets respectively. The results are shown in Table 11.

In Table 11, under normal light conditions, compared with the conventional YOLO series algorithms and multimodal remote sensing image object detection algorithms, the MMFDet algorithm can maintain high detection accuracy. Under low-light conditions, the detection accuracy of the MMFDet algorithm is substantially enhanced by 16.1% over the baseline algorithm YOLOv8n. Compared with the multimodal remote sensing image object detection algorithms YOLOrs and SuperYOLO, the detection accuracy of MMFDet algorithm is enhanced by 4.5% and 5.9%, respectively. It can be seen that the MMFDet algorithm is able to maintain a high detection accuracy under normal light conditions, and the detection accuracy under low-light conditions is greatly enhanced.

4.3.3. Visualization results analysis

This paper presents visualization results of object detection on both the VEDAI dataset and the Drone Vehicle dataset, illustrated in Figs. 8 and 9, respectively.

Fig. 8 shows the results of small object detection by YOLOv8n and MMFDet. In Fig. 8, we use red arrows to denote the objects missed by the YOLOv8n algorithm and blue arrows to denote the wrongly detected objects. By comparing the visualization results in Fig. 8, we can find that there are a large number of small objects in all three

Table 7
Comparison of detailed detection results of different algorithms using the VEDAI dataset.

Method	Car	Pickup	Camping	Truck	Other	Tractor	Boat	Van	mAP/%	Params.
YOLOv3	84.6	72.7	67.1	62.0	43.0	65.2	37.1	58.3	61.3	61.5M
YOLOv4	85.4	72.8	72.3	62.8	48.9	68.9	34.2	54.6	62.5	52.5M
YOLOv5s	80.8	68.4	69.0	54.7	46.7	64.3	24.2	45.9	56.8	7.1M
YOLOv8n	84.3	59.1	71.5	59.6	63.5	48.5	42.3	76.6	63.2	**3.0M**
YOLOv8s	79.8	65.1	65.5	57.3	63.2	57.9	42.5	46.0	59.7	11.1M
YOLOv8m	83.1	64.8	66.9	63.0	60.9	63.1	36.1	60.2	62.2	25.8M
YOLOv8l	84.2	68.8	61.1	68.5	49.2	71.8	40.7	60.1	63.1	43.6M
YOLOv8x	86.1	74.4	67.3	**77.0**	57.1	68.2	41.8	66.3	67.3	68.1M
YOLOrs	80.6	72.8	55.3	55.8	44.4	56.7	50.3	57.3	59.2	20.2M
SuperYOLO	81.7	77.9	64.9	71.2	31.9	77.9	62.8	73.5	67.7	4.8M
MMFDet	**88.3**	**78.5**	**81.6**	59.8	**63.5**	**86.2**	**76.0**	**88.3**	**77.9**	3.5M

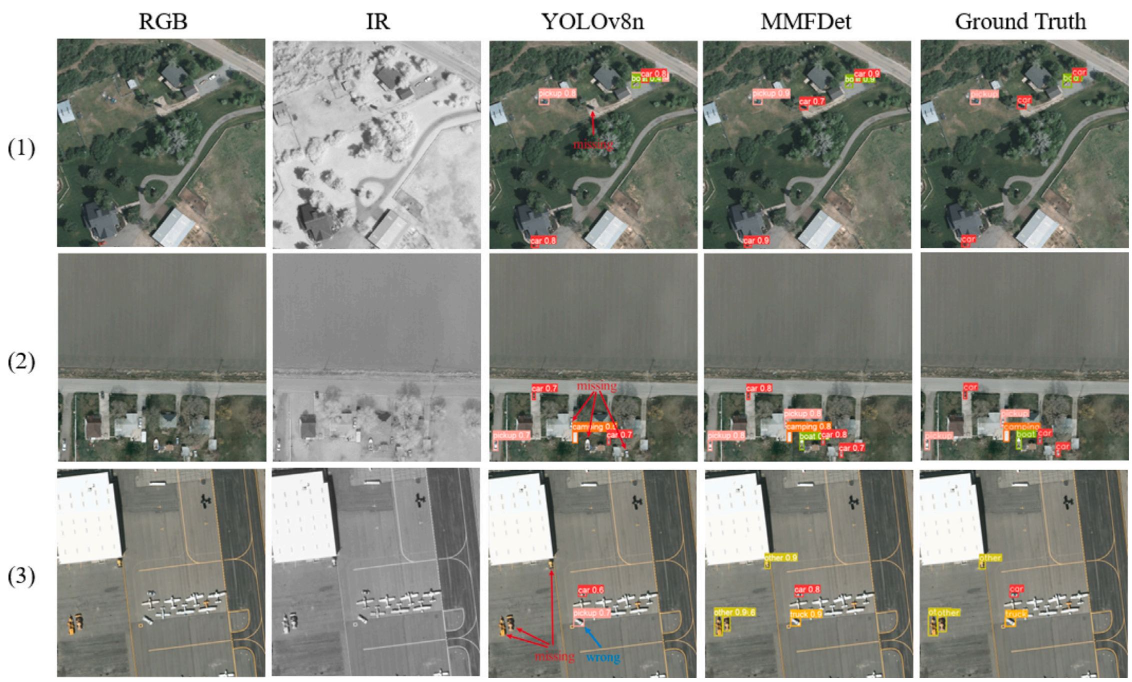

Fig. 8. Visualization detection results on the VEDAI dataset.

Table 8
Ablation studies on the Drone Vehicle dataset.

Method	mAP/%
Baseline	71.4
Baseline+MDCM	81.8
Baseline+HFSHM	71.8
Baseline+MDCM+HFSHM	82.1
Baseline+MDCM+MBE	82.4
Baseline+MDCM+MBE+HFSHM	**82.5**

Table 9
Comparison of detection results of different algorithms on Drone Vehicle dataset.

Method	mAP/%	FNR/%	Params.
YOLOv3	71.2	30.6	61.5M
YOLOv4	72.5	30.1	52.5M
YOLOv5s	70.6	33.2	7.1M
YOLOv8n	71.4	32.5	**3.0M**
YOLOv8s	74.2	30.1	11.1M
YOLOv8m	75.9	29.0	25.8M
YOLOv8l	76.4	27.1	43.6M
YOLOv8x	76.2	28.8	68.1M
YOLOrs	75.4	22.6	20.2M
SuperYOLO	77.9	22.8	4.8M
MMFDet	**82.5**	**21.7**	3.5M

Table 10
Comparison of detailed detection results of different algorithms using the Drone Vehicle dataset.

Method	Freight car	Car	Truck	Bus	Van	mAP/%	Params.
YOLOv3	89.2	69.1	93.5	52.8	51.3	71.2	61.5M
YOLOv4	90.1	69.9	93.3	55.7	53.3	72.5	52.5M
YOLOv5s	89.8	65.3	91.8	55.4	50.8	70.6	7.1M
YOLOv8n	90.9	66.8	91.9	54.7	52.7	71.4	**3.0M**
YOLOv8s	92.1	70.7	93.3	60.0	54.8	74.2	11.1M
YOLOv8m	92.7	72.4	84.2	62.1	58.3	75.9	25.8M
YOLOv8l	92.9	72.6	95.0	62.9	58.5	76.4	43.6M
YOLOv8x	92.8	72.2	95.3	62.8	57.9	76.2	68.1M
YOLOrs	97.0	69.4	95.5	57.6	57.2	75.3	20.2M
SuperYOLO	97.8	71.2	95.0	62.5	62.8	77.9	4.8M
MMFDet	**98.4**	**80.7**	**97.2**	**66.7**	**69.3**	**82.5**	3.5M

sets of images. For the pictures in (1) and (2), YOLOv8n cannot detect the small objects in the pictures when they are interfered by the background information. And for (3), the YOLOv8n algorithm wrongly detects and misses the objects even if the interference of background noise is weak. MMFDet, on the other hand, can effectively reduce the interference of background noise on small objects through the design of convolutional kernel parameters in the multi-branch feature extraction structure and the processing of high-level features by utilizing the high-level feature split hybrid module, which can effectively detect the small

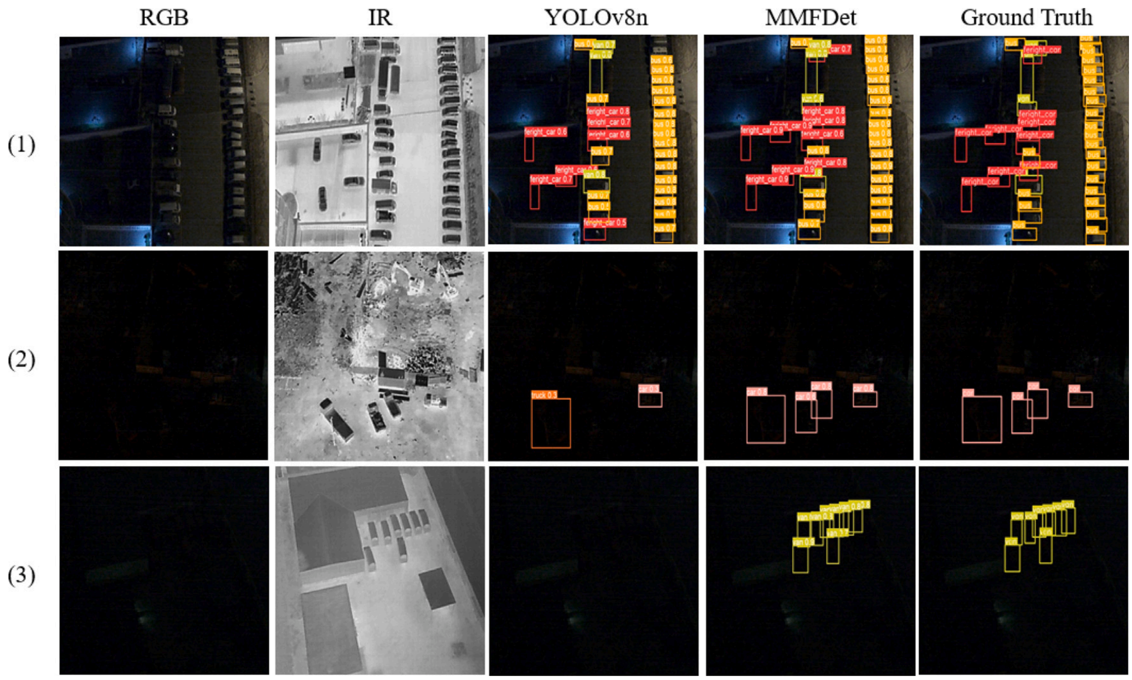

Fig. 9. Visualization detection results on the Drone Vehicle dataset.

Table 11
Comparison of experimental results under normal light conditions and low light conditions.

Method	Normal light (mAP/%)	Low-light (mAP/%)
YOLOv3	77.3	70.2
YOLOv4	77.4	72.7
YOLOv5s	75.7	66.8
YOLOv8n	72.6	67.5
YOLOv8s	74.5	70.7
YOLOv8m	76.5	71.8
YOLOv8l	76.6	73.7
YOLOv8x	77.3	72.5
YOLOrs	79.3	79.1
SuperYOLO	76.0	77.7
MMFDet	**80.4**	**83.6**

objects detection in the three sets of images, and reduce the cases of false positives and missed detections.

Fig. 9 shows the object detection results of YOLOv8n and MMFDet in low-light environments. From (1) in Fig. 9, it can be seen that the YOLOv8n algorithm is able to detect the objects in images with relatively good lighting conditions, but there are more misdetections and missed detections. And the detection results of MMFDet are almost the same as Ground Truth. According to the visualization results of (2) and (3), it can be seen that YOLOv8n can hardly detect the objects in images with low lighting conditions, and even if it is able to locate some objects, the confidence level is very low and it is unable to identify the category of the object. In contrast, MMFDet better detects the objects in images with low lighting conditions by combining the texture and contour information of the RGB modality with the thermal radiation information of the IR modality through the MDCM module. It is proved that the MDCM module can effectively improve the ability of detecting objects in images with low lighting environment.

5. Conclusions

This paper presents a novel differential multimodal fusion algorithm for remote sensing image object detection, utilizing multi-branch feature extraction. The algorithm demonstrates proficient positioning and recognition of small objects and objects in low-light environments. Through a comparative analysis using the VEDAI and Drone Vehicle datasets, the following conclusions were drawn in this study:

(1) Compared with the baseline algorithm, our proposed approach yields substantial enhancements, with improvements of 14.7% and 11.1% in mAP values on the VEDAI and Drone Vehicle datasets, respectively. Moreover, when contrasted with traditional and other advanced algorithms, our method exhibits noteworthy advancements in mAP values.

(2) Visual inspection of detection outcomes between the baseline algorithm and our proposed approach reveals that our method significantly bolsters the detection accuracy of small objects and objects in low-light environments. This improvement is attributed to the astute design of the network structure and the fusion of features extracted from RGB and IR images.

While our algorithm has considerably enhanced detection accuracy, there remains potential for further optimization, particularly in reducing the model size. Future research endeavors should focus on exploring more streamlined structures to strike a balance between accuracy and model complexity.

CRediT authorship contribution statement

Wenqing Zhao: Conceptualization, Methodology, Validation, Investigation, Writing – review & editing, Supervision, Funding acquisition. **Zhenhuan Zhao:** Methodology, Software, Validation, Investigation, Data curation, Writing – original draft. **Minfu Xu:** Validation, Investigation, Writing – review & editing, Supervision. **Yingxue Ding:** Methodology, Validation, Formal analysis, Writing – review & editing, Supervision. **Jiaxiao Gong:** Methodology, Validation, Investigation, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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